

MATHEMATICS

TARGET : JEE (Main) 2024

CHAMP-2024

REVISION DPP

NO. 12

Syllabus:- Application of Derivatives

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The sum of absolute maximum and absolute minimum values of the function $f(x) = |2x^2 + 3x - 2| + \sin x \cos x$ in the interval $[0, 1]$ is :
 - $3 + \frac{\sin(1)\cos^2(1/2)}{2}$
 - $3 + \frac{1}{2}(1 + 2\cos(1))\sin(1)$
 - $5 + \frac{1}{2}(\sin(1) + \sin(2))$
 - $2 + \sin\left(\frac{1}{2}\right)\cos\left(\frac{1}{2}\right)$
- If $ax^2 + \frac{b}{x} \geq c \forall x \in \mathbb{R}^+$, where $a > 0$ and $b > 0$, then
 - $27ab^2 \geq 4c^3$
 - $27ab^3 \geq 4c^3$
 - $27a^2b^2 \geq 4c^3$
 - $27a^3b \geq 4c^3$
- The function $f(x) = x - \ln |2x + 1|$, $x \in \left(-100, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, \frac{1}{2}\right)$ is decreasing in interval
 - $\left(-\frac{1}{2}, \frac{1}{2}\right)$
 - $\left(-100, -\frac{1}{2}\right)$
 - $\left(-\frac{1}{2}, 0\right)$ only
 - $\left(0, \frac{1}{2}\right)$ only
- If the equation $x^3 - 9x^2 + 24x + k = 0$ has three real distinct roots then set of values of k is
 - $(-\infty, -20)$
 - $(16, 20)$
 - $(-20, -16)$
 - $(20, \infty)$
- The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point $(1, 1)$ and the coordinate axes lies in the first quadrant. If its area is 2 square units then $b =$
 - 1
 - 3
 - 3
 - 1
- Let $f(x) = x^3 + ax^2 + bx + 5 \sin^2 x$ be an increasing function in the set of real numbers $\mathbb{R}$. Then a and b satisfy the condition –
 - $a^2 - 3b - 15 > 0$
 - $a^2 - 3b + 15 > 0$
 - $a^2 - 3b + 15 < 0$
 - $a > 0$ and $b > 0$
- The function $f(x) = \sin^4 x + \cos^4 x$ increases if
 - $0 < x < \frac{\pi}{8}$
 - $\frac{\pi}{4} < x < \frac{3\pi}{8}$
 - $\frac{3\pi}{8} < x < \frac{5\pi}{8}$
 - $\frac{5\pi}{8} < x < \frac{3\pi}{4}$
- On which of the following intervals is the function $x^{100} + \sin x - 1$ decreasing –
 - $(0, \pi/2)$
 - $(0, 1)$
 - $(\pi/2, \pi)$
 - none
- If $f(x) = \left(\frac{3}{5}\right)^x + \left(\frac{4}{5}\right)^x - 1$, $x \in \mathbb{R}$, then the equation $f(x) = 0$ has
 - no solution
 - one solution
 - two solutions
 - more than two solutions

10. Let $f(x) = \min(4x + 1, x + 2, -2x + 4)$, $x \in \mathbb{R}$ then the maximum value of $f(x)$ is
- (1) $\frac{1}{3}$ (2) $\frac{1}{2}$ (3) $\frac{2}{3}$ (4) $\frac{8}{3}$
11. Let k and K be the minimum and the maximum values of the function $f(x) = \frac{(1+x)^{0.6}}{1+x^{0.6}}$ in $[0, 1]$ respectively, then their ordered pair (k, K) is equal to:
- (1) $(2^{-0.4}, 1)$ (2) $(2^{-0.4}, 2^{0.6})$ (3) $(2^{-0.6}, 1)$ (4) $(1, 2^{0.6})$
12. If $A > 0$, $B > 0$ and $A + B = \frac{\pi}{6}$, then the minimum value of $\tan A + \tan B$ is
- (1) $2 - \sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $\sqrt{3} - \sqrt{2}$ (4) $4 - 2\sqrt{3}$
13. Let $f(x) = |(x-1)(x^2 - 2x - 3)| + x - 3$, $x \in \mathbb{R}$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval $(0, 4)$, then $m + M$ is equal to ____.
- (1) 2 (2) 6 (3) 0 (4) 3
14. Let $f(x) = \begin{cases} |x-1| + a & \text{if } x \leq 1 \\ 2x+3 & \text{if } x > 1 \end{cases}$. If $f(x)$ has a local minimum at $x = 1$ then
- (1) $a > 5$ (2) $0 < a \leq 5$ (3) $a \leq 5$ (4) $a = 5$
15. The equation $4x^5 + 5x^4 - 10x^2 - 20x + 40 = 0$ has
- (1) No real root (2) one real root (3) three real roots (4) Five real roots
16. Let $f(x) = [1 - x^2]$, $x \in \mathbb{R}$, where $[.]$ is the greatest integer function, then :
- (1) f is increasing (2) $x = 0$ is the point of maxima of f
 (3) f is continuous at $x = 0$ (4) f is decreasing
17. The maximum value of $f(x) = 2\sin x + \sin 2x$, in the interval $\left[0, \frac{3}{2}\pi\right]$, is
- (1) $\sqrt{2} + 1$ (2) $2\sqrt{3}$ (3) $\frac{3\sqrt{3}}{2}$ (4) $\sqrt{3}$
18. The sum of the absolute minimum and the absolute maximum values of the function $f(x) = |3x - x^2 + 2| - x$ in the interval $[-1, 2]$ is :
- (1) $\frac{\sqrt{17} + 3}{2}$ (2) $\frac{\sqrt{17} + 5}{2}$ (3) 5 (4) $\frac{9 - \sqrt{17}}{2}$
19. Let $f(x) = x^2 e^{-2x}$ ($x > 0$). Then $e^2 f(x)$ has maximum value equal to –
- (1) 0 (2) 1 (3) 2 (4) 3
20. A point on the hypotenuse of a right angled triangle is at distance a and b from the sides of the triangle. If the minimum length of the hypotenuse is $(a^{k/3} + b^{k/3})^{3/2}$ then $k =$
- (1) 1 (2) 2 (3) 4 (4) 5

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. The function $f(x) = (x - 2)^3 (x - 1)^2$ has relative minimum at $x =$
22. If $f(x) = \begin{cases} \ln x, & x \leq e \\ -3x + k, & x > e \end{cases}$ then the number of non negative integral values of k so that $f(x)$ has global maximum at $x = e$, is
23. Consider a cuboid of sides $2x$, $4x$ and $5x$ and a closed hemisphere of radius r . If the sum of their surface areas is a constant k , then the ratio $15x/r$, for which the sum of their volumes is maximum, is :
24. The number of values of x where the function $f(x) = \cos x + \cos(\sqrt{2}x)$ attains its maximum value is
25. Ship A is 6 kilometre west of a point O and moving east at 3 Kilometre per hour, ship B is 8 kilometre North of O and moving south at 4 kilometre per hour. They are nearest to one another, when time in hours equals

ANSWER KEY_DPP-12

1.	(2)	2.	(1)	3.	(1)	4.	(3)	5.	(2)	6.	(3)	7.	(2)
8.	(4)	9.	(2)	10.	(4)	11.	(1)	12.	(4)	13.	(4)	14.	(3)
15.	(2)	16.	(2)	17.	(3)	18.	(1)	19.	(2)	20.	(2)	21.	(01.4)
22.	(10)	23.	(19)	24.	(1)	25.	(2)						